Classifications

Linear: $\alpha_1 x_1(t) + \alpha_2 x_2(t) = \alpha_1 y_1(t) + \alpha_2 y_2(t)\ \forall\ \alpha_1, \alpha_2$
To prove, sum first, then transform, and prove equal to, transform first then sum.
Energy:

$$E = \int_{t_1}^{t_2} |x(t)|^2 dt = \int_{t_1}^{t_2} (x_{Re}^2(t) + x_{Im}^2(t)) dt$$

for total energy, $t_1 = -\infty, t_2 = \infty$.
Power:

$$P_{avg} = \frac{1}{t_2 - t_1} E = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} (x_{Re}^2(t) + x_{Im}^2(t)) dt$$

(In discrete time, the denominator is $n_2 - n_1 + 1$). For total, both limits tend to ∞.
Period: $\exists T \neq 0 | x(t) = x(t+T)\ \forall t \implies$ periodic.
Minimum T that fulfills conditions is T_0, fundamental period. For two added signals, $T = LCM(T_1, T_2)$.
$T(\sin(\omega t)) = \frac{2\pi}{\omega}$
Parity:
Even (+): $x(t) = x(-t)$
Odd (-): $x(t) = -x(-t)$

$$x = even + odd = \frac{x(t) + x(-t)}{2} + \frac{x(t) - x(-t)}{2}$$

Complex Exponential:

$$\begin{aligned} x(t) = Ce^{\alpha t} &= |C| \cdot e^{\sigma t} \cdot e^{j(\omega t + \phi)} \\ &= |C| \cdot e^{\sigma t} \cdot (\cos(\omega t + \phi) + \sin(\omega t + \phi)) \end{aligned}$$

$\omega = 2\pi/T$ is the fundamental frequency.

$$P(t) = |C|^2 w^{2\sigma t}$$
$$E_\infty = \infty, P_\infty = C^2$$

HRCEs: $x_k(t) = e^{jk\omega_0 t}$

Transformations

Time Shift:$x(t) \to x(t - t_0)$.Moves signal right by t_0.
Time Reversal:$x(t) \to x(-t)$.Reflects signal about y-axis.
Time Scale:$x(t) \to x(\alpha t)$.Widens signal by $\frac{1}{\alpha}$.

Complex Exponential

DTCE: $x[n] = |C| \cdot e^{\sigma n} \cdot e^{j\omega n + \phi}$
Has exponential envelope $|C|e^{\sigma t}$.

$$N_0 = LCM\left(T = \frac{2\pi}{\omega}, 1\right)$$

For sum, $N = LCM(N_1, N_2)$.
If no π in ω, or ω is irrational, the signal is not periodic. $0 \leq \omega < 2\pi$.
Speed up for $\omega \approx \pm\pi$, slow at borders, that is for $k = \lfloor \frac{N}{2} \rfloor or \lceil \frac{N}{2} \rceil$.
DTHRCEs: $x_k[n] = e^{jk(\frac{2\pi}{N})n}$
Only N distinct members in the HRCE: $k = 0 \dots N-1$.

Special Signals

DT:
$\delta[n] = 0\ \forall\ n \neq 0,$ else 1.
$u[n] = \sum_{k=0}^{\infty} \delta[n-k] = 1\ \forall\ n \geq 0$
$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$
CT:
$\int_{-\infty}^{\infty} \delta(t) = 1$

System Classifications

Series:$x \to [S1] \to [S2] \to y$
Parallel:

$$\begin{array}{ll} & \uparrow \to [S1] \to \downarrow \\ x & \qquad + \to y \\ & \downarrow \to [S2] \to \uparrow \end{array}$$

Feedback:

$$\begin{array}{ll} x \to + \to [S1] \to y \\ \qquad \uparrow \leftarrow [S2] \leftarrow \downarrow \end{array}$$

Properties

Memory: if $y(t) = f(x(T))|T \neq t_0$, the system has memory.
Invertible: $x \to [S] \to [S^{-1}] \to x$

$$\forall x_1, x_2 | x_1 \neq x_2 \implies y_1 \neq y_2$$

Can be proven if x as a function of y is defined across all t.
Causal: $y(t) = \sum_{T=-\infty}^{t} f(x(T))$. No dependence on future values of t.
Stable: $|x(t)| < B \implies |y(t)| < M$ for finite B, M.
Time Invariant: $x(t) \to y(t) \implies x(t-t_0) \to y(t-t_0)$. Scaling, or having a function of t outside, like $(tx(t))$, will **NOT** be time invariant.

SinCos, etc.

$$\cos(x) = \frac{e^{jx} + e^{-jx}}{2} \ , \ \sin(x) = \frac{e^{jx} - e^{-jx}}{2j} \ , \ \omega = \frac{2\pi}{T} = \frac{2\pi}{N}$$

$\mathbf{F(x)} = X(e^{j\omega})$ has a period of 2π
$x = |x|e^{j\angle \underline{x}}$

Classifications

Memoryless: $h[k] = 0 = h(t) \ \forall \ k, t \neq 0$ **OR** $y(t) = f(x(T))|T \neq t \implies$ memory
Invertible: $\exists h_I | h * h_I = \delta$ **OR** $H(j\omega) \neq 0$ **OR** $H(e^{j\omega}) \neq 0$ **OR** $\forall x_1, x_2 | x_1 \neq x_2 \implies y_1 \neq y_2$
Causal: $h[k] = h(t) = 0 \ \forall \ k, t < 0$ **OR** $y(t) = \sum_{T=-\infty}^{t} f(x(T))$. No dependence on future values of t.
Stable: $\sum_{-\infty}^{\infty} |h[k]| < \infty$,$\int_{-\infty}^{\infty} h(\tau) d\tau < \infty$ **OR** H is defined **OR** $|x(t)| < B \ \forall \ t \implies |y(t)| < M \ \forall \ t$ for bounded B, M

Fourier

$$y(t) = e^{j\omega t} * h(t) = e^{j\omega t} \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau = e^{j\omega t} |H(j\omega)| e^{j\angle \underline{H}(j\omega)} = \sum_{\omega} \mathbf{x(e^{j\omega t})H(j\omega)}$$

$$a_k = \frac{1}{T} \int_{<T>} x(t) e^{-jk\omega_0 t} dt = \frac{X(j\omega)}{T} = \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk\frac{2\pi}{N}n} = \frac{X(e^{jk\omega_0})}{N}$$

$$x(t) = \sum_{-\infty}^{\infty} a_k e^{jk\omega_0 t} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \ , \ x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n} = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

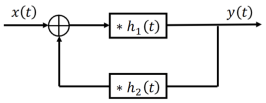
$$X(j\omega) = 2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - \frac{2\pi K}{T}) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \ , \ X(e^{j\omega}) = 2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - \frac{2\pi K}{N}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$\sum_{m=-\infty}^{n} x[m] \to \frac{1}{1 - e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j0}) \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

Real $x(t)$: $a_k = a^*_{-k}$
Even $x(t)$: $a_k = a_{-k}$
Duality: $\mathbf{F}\{X(t)\} = 2\pi \cdot x(-\omega)$
Convergence: $E_\infty < \infty$

LTI

$x(t) \to [H_1] \to [H_2] \to y(t) = x(t) \to [H_1 \cdot H_2] \to y(t)$



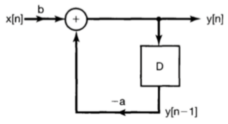
$Y(j\omega) = H_1(j\omega)(X(j\omega) + H_2(j\omega)Y(j\omega)) = \frac{H_1(j\omega)}{1 - H_1(j\omega)H_2(j\omega)}$
$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$

Diff

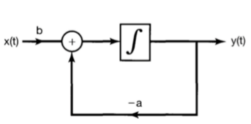
Nth order linear constant diffEq:

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k] \implies H(e^{j\omega}) = \frac{\sum_{k=0}^{M} b_k e^{-jk\omega}}{\sum_{k=0}^{N} a_k e^{-jk\omega}}$$

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k} \implies H(j\omega) = \frac{\sum_{k=0}^{M} b_k (j\omega)^k}{\sum_{k=0}^{N} a_k (j\omega)^k}$$



(a) $y[n] + ay[n-1] = bx[n]$



(b) $\frac{dy(t)}{dt} + ay(t) = bx(t)$

Property	Aperiodic Signal	Fourier Transform	Coefficients
Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$	$a \times a_k + b \times b_k$
Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0}X(j\omega)$	$a_k e^{-j\omega_0 t_0}$
Frequency Shifting	$e^{j\omega_0 t}x(t)$	$X(j(\omega - \omega_0))$	a_{k-n}
Conjugation	$x^*(t)$	$X^*(-j\omega)$	a^*_{-k}
Time Reversal	$x(-t)$	$X(-j\omega)$	a_{-k}
Time/Frequency Scaling	$x(at)$	$\frac{1}{\lvert a\rvert} X\left(\frac{j\omega}{a}\right)$	a_k (with scaled period)
Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$	$a_k b_k$
Multiplication	$x(t)y(t)$	$\frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\theta)Y(j(\omega-\theta))d\theta$	convolution of a_k and b_k
Differentiation in Time	$\frac{d}{dt}x(t)$	$j\omega X(j\omega)$	$jk\omega_0 a_k$
Integration	$\int_{-\infty}^{t} x(\tau)d\tau$	$\frac{1}{j\omega}X(j\omega) + \pi X(0)\delta(\omega)$	$\frac{a_k}{jk\omega_0}$ (if $a_0 = 0$)
Differentiation in Frequency	$tx(t)$	$j\frac{d}{d\omega}X(j\omega)$	$-j\frac{d}{d\omega_0}a_k$
Conjugate Symmetry	$x(t)$ real	$\begin{cases} X(j\omega) = X^*(-j\omega) \\ \mathrm{Re}[X(j\omega)] = \mathrm{Re}[X(-j\omega)] \\ \mathrm{Im}[X(j\omega)] = -\mathrm{Im}[X(-j\omega)] \end{cases}$	$a_k = a^*_{-k}$
Even/Odd Decomposition	$x_e(t), x_o(t)$	$\mathrm{Re}[X(j\omega)], j\mathrm{Im}[X(j\omega)]$	$\mathrm{Re}[a_k], \mathrm{Im}[a_k]$
Parseval's Theorem	$\int_{-\infty}^{\infty}\lvert x(t)\rvert^2 dt$	$= \frac{1}{2\pi}\int_{-\infty}^{\infty}\lvert X(j\omega)\rvert^2 d\omega$	$\sum_{k=-\infty}^{\infty}\lvert a_k\rvert^2$

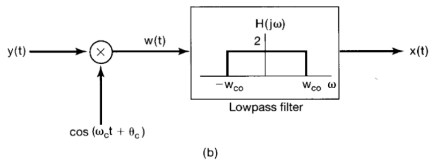
Signal	Fourier Transform	FS Coefficients (if periodic)
$\sum_{-\infty}^{\infty} a_k e^{jk\omega_0 t}$	$2\pi\sum_{-\infty}^{\infty} a_k\delta(\omega - k\omega_0)$	a_k
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$	$a_1 = 1$, else 0
$\cos(\omega_0 t)$	$\pi[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)]$	$a_1 = a_{-1} = \frac{1}{2}$, else 0
$\sin(\omega_0 t)$	$\frac{\pi}{j}[\delta(\omega-\omega_0) - \delta(\omega+\omega_0)]$	$a_1 = -a_{-1} = \frac{1}{2j}$, else 0
$x(t) = 1$	$2\pi\delta(\omega)$	$a_0 = 1$, $a_k = 0$, $k \neq 0$
Square Wave ($\lvert t\rvert < T_1$), $x(t) = x(t+T)$	$\sum_{-\infty}^{\infty}\frac{2\sin(k\omega_0 T_1)}{k}\delta(\omega - k\omega_0)$	$a_k = \frac{\sin(k\omega_0)T_1}{k\pi}$
$\sum_{n=-\infty}^{\infty}\delta(t - NT)$	$\frac{2\pi}{T}\sum_{k=-\infty}^{\infty}\delta(\omega - \frac{2\pi k}{T})$	$a_k = \frac{1}{T}$
Rectangular pulse (1 for $\lvert t\rvert < T_1$)	$\frac{2\sin(\omega T_1)}{\omega}$	—
$sin(Wt)/\pi t$	1 for $\lvert\omega\rvert < W$	—
$\delta(t)$	1	—
$u(t)$	$\frac{1}{j\omega} + \pi\delta(\omega)$	—
$\delta(t - t_0)$	$e^{-j\omega t_0}$	—
$e^{-at}u(t)$, $\mathrm{Re}\{a\} > 0$	$\frac{1}{a+j\omega}$	—
$t^{(n-1)}/(n-1)! \cdot e^{-at}u(t)$, $\mathrm{Re}\{a\} > 0$	$\frac{1}{(a+j\omega)^n}$	—

Property	Aperiodic Signal	Fourier Transform	Coefficients
Linearity	$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	$aa_k + bb_k$
Time Shifting	$x[n - n_0]$	$e^{-j\omega n_0}X(e^{j\omega})$	$a_k e^{-j\omega_0 n_0}$
Frequency Shifting	$e^{j\omega_0 n}x[n]$	$X(e^{j(\omega-\omega_0)})$	$a_{k-m} mod(N)$
Conjugation	$x^*[n]$	$X^*(e^{-j\omega})$	a^*_{-k}
Time Reversal	$x[-n]$	$X(e^{-j\omega})$	$a_{-k} mod(N)$
Time Expansion	$x_a[n] = x[n/k], n =$ multiple of k	$X(e^{j\omega})$	$a_{k/k'}$ (nonzero at multiples)
Convolution	$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	$a_k * b_k$
Multiplication	$x[n]y[n]$	$\frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$	$a_k * b_k$ in frequency domain
Difference in Time	$x[n] - x[n-1]$	$(1 - e^{-j\omega})X(e^{j\omega})$	$a_k - a_{k-1}$
Accumulation	$\sum_{k=-\infty}^{n} x[k]$	$\frac{1}{1-e^{-j\omega}}X(e^{j\omega}) + \pi X(e^{j\omega})\sum\delta(\omega - 2\pi k)$	$\frac{a_k}{1-e^{-j\omega_0}}$ (approximate form)
Differentiation in Frequency	$nx[n]$	$j\frac{dX(e^{j\omega})}{d\omega}$	$j\frac{da_k}{d\omega_0}$ or $-jka_k$
Conjugate Symmetry	$x[n]$ real	$\begin{cases} X(e^{j\omega}) = X^*(e^{-j\omega}) \\ \mathrm{Re}[X(e^{j\omega})] = \mathrm{Re}[X(e^{-j\omega})] \\ \mathrm{Im}[X(e^{j\omega})] = -\mathrm{Im}[X(e^{-j\omega})] \end{cases}$	$a_k = a^*_{-k}$
Even/Odd Decomposition	$x[n] = x_e[n] + x_o[n]$	$\mathrm{Re}[X(e^{j\omega})], j\mathrm{Im}[X(e^{j\omega})]$	$\mathrm{Re}[a_k], \mathrm{Im}[a_k]$
Parseval's Theorem	$\sum_{n=-\infty}^{\infty}\lvert x[n]\rvert^2$	$= \frac{1}{2\pi}\int_{-\pi}^{\pi}\lvert X(e^{j\omega})\rvert^2 d\omega$	$\sum_{k=-\infty}^{\infty}\lvert a_k\rvert^2$

Signal	Fourier Transform	FS Coefficients (if periodic)
$\sum a_k e^{j\frac{2\pi}{N}kn}$	$2\pi\sum a_k\delta(\omega - \frac{2\pi k}{N})$	a_k
$e^{j\omega_0 n}$	$2\pi\sum_r\delta(\omega - \omega_0 - 2\pi r)$	$a_1 = 1$, $k = m \pm N, m \pm 2N$
$\cos(\omega_0 n)$	$\pi\sum_r[\delta(\omega-\omega_0-2\pi r) + \delta(\omega+\omega_0-2\pi r)]$	$a_{\pm k} = \frac{1}{2}$ for $k = \pm m$
$\sin(\omega_0 n)$	$\frac{\pi}{j}\sum_r[\delta(\omega-\omega_0-2\pi r) - \delta(\omega+\omega_0-2\pi r)]$	$a_1 = -a_{-1} = \frac{1}{2j}$
$x[n] = 1$	$2\pi\sum_r\delta(\omega - 2\pi r)$	$a_k = 1, k = \pm mN$
$x[n] = 1$ for $\lvert n\rvert < N_1$, $x[n+N] = x[n]$	$2\pi\sum_{k=-\infty}^{\infty} a_k\delta(\omega - 2\pi k/N)$	$a_{mN} = \frac{2N_1+1}{N}, a_{!mN} = \frac{\sin((2\pi k/N)(N_1+1/2))}{N\sin(2\pi k/2N)}$
$\sum_{k=-\infty}^{\infty}\delta[n - kN]$	$\frac{2\pi}{N}\sum_{k=-\infty}^{\infty}\delta(\omega - 2\pi k/N)$	$a_k = \frac{1}{N}$
$x[n] = 1$ for $\lvert n\rvert < N_1$	$\frac{\sin(\omega(N_1+1/2))}{\sin(\omega/2)}$	—
$\sin(Wn)/\pi n, 0 < W < \pi$	1 for $\lvert\omega\rvert < W$, periodic	—
$\delta[n]$	1	—
$u[n]$	$\frac{1}{1-e^{-j\omega}} + \pi\sum\delta(\omega - 2\pi k)$	—
$\delta[n - n_0]$	$e^{-j\omega n_0}$	—
$a^n u[n], \lvert a\rvert < 1$	$\frac{1}{(1-ae^{-j\omega})}$	—
$(n+1)a^n u[n], \lvert a\rvert < 1$	$\frac{1}{(1-ae^{-j\omega})^2}$	—
$\frac{(n+r-1)!}{n!(r-1)!}a^n u[n], \lvert a\rvert < 1$	$\frac{1}{(1-ae^{-j\omega})^r}$	—
$\sum_{k=-\infty}^{n} x[n], a_0 = 0$	—	$\frac{a_k}{1-e^{-jk(2\pi/N)}}$

Amplitude and Frequency Modulation

- $c(t) = e^{j\omega_c t + \theta_c} \implies Y(j\omega) = X(j\omega - j\omega_c)$.
  Note that $x(t) = y(t)e^{-j\omega_c t}$
  and $y(t) = x(t)\cos(\omega_c t) + jx(t)\sin(\omega_c t)$
- $c(t) = \cos(\omega_c t + \theta_c) \implies Y(j\omega) = \frac{1}{2}[X(j\omega - j\omega_c) + X(j\omega + \omega_c)]$
  $c(t) = \sin(\omega_c t + \theta_c) \implies Y(j\omega) = \frac{1}{2j}[X(j\omega - j\omega_c) - X(j\omega + \omega_c)]$
  Note that $x(t)$ is only recoverable if $\omega_M < \omega_{co} < 2\omega_c - \omega_M$, where ω_M is the bandwidth of $x(j\omega)$
  Demodulation: $w(t) = x(t)\cos^2(\omega_c t)$



(b)

Sampling

Sampling is done using an impulse train, that is:

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

$$P(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s)$$

$$X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j\omega - jk\omega_s)$$

For no overlap: $\omega_s = \frac{2\pi}{T} > 2\omega_M$